Tran Trung Duc

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AI engineer building **production agentic systems and LLM applications** that run real business operations. I design multi-agent workflows, RAG and retrieval pipelines, and the **evaluation frameworks** that make model output reliable enough to ship — keeping deterministic computation in Python and letting the LLM handle language. Background spans deep-learning research through backend engineering down to embedded firmware, with a bias toward shipping and measuring in production.

PROFESSIONAL EXPERIENCE

Maxflow AI Ho Chi Minh City, Vietnam *Feb 2026 – Present*
AI Automation / Agent Builder

- **Designed and shipped multi-agent systems for enterprise clients** across B2B sales automation, inventory logistics, education & event operations, email pipelines, and HR workflows — master routing agents delegating to specialist sub-agents with tool-calling, memory, and state across long-running processes
- **Built RAG and retrieval pipelines**: hybrid **BM25 keyword + embedding semantic search** over vector stores to ground and fact-check LLM output against source documents, cutting false positives on document-grounded tasks
- **Engineered reliability into LLM output**: strict JSON schemas, structured validation, guardrails, fallback and retry logic, plus **Playwright** automated E2E suites and transcript-based debugging to trace failures across model, orchestration, and API layers
- **Optimised inference latency and cost**: routed high-volume steps to cheap/fast models and reserved larger models for hard reasoning; tuned context size, cached retrieved context, streamed responses, and handled timeouts, retries, and rate limits under load
- **Integrated AI into backend and enterprise systems**: REST/**FastAPI** services, event-driven triggers, scheduled jobs, webhook receivers, a custom **MCP** CRM connector, external gateways (SMS, OA, SMTP), 12-table relational schemas, and HTML/JS mini-app frontends
- **Deployed and documented for reuse**: **Docker** and **Cloud Run** with CI/CD; reusable agent-building playbooks and architecture notes. Used **Claude Code** daily as the primary development harness

SNOW Vietnam Corporation Ho Chi Minh City, Vietnam *May 2025 – Oct 2025*
Data Labeling / Product Reviewer (Intern) — AI & Computer Vision Features

- Evaluated AI-generated image and video features (AR Filters, computer-vision effects) shipped to a large consumer user base; designed test cases covering **edge cases** under low light, occlusion, and unusual input
- Documented reproducible defect patterns for the CV research team, verified fixes, and tracked regressions across releases — tightening the feedback loop between QA and model development

ITR VN (ITR Vietnam) Ho Chi Minh City, Vietnam *Nov 2024 – Feb 2025*
Artificial Intelligence Intern

- **Reproduced and trained the PPG2BP-Net deep-learning architecture** (**PyTorch**, CNN-LSTM, Transformer) on MIMIC-III for cuffless blood pressure estimation, reaching clinical-grade accuracy: SBP MAE 9.76 ± 8.99 mmHg, DBP MAE 5.25 ± 5.26 mmHg, MAP MAE 6.26 ± 5.89 mmHg
- **Built end-to-end ML data pipelines**: raw waveform ingestion, cleaning, preprocessing, feature extraction, training, and evaluation; implemented custom loss functions and MAE/RMSE metrics, analysing bias–variance tradeoffs to prevent overfitting
- **Cut processing latency $\sim$40%** by vectorising the hot path with NumPy; wrote unit tests for data-processing functions and structured the work as a reproducible research framework

SELECTED PROJECTS

Multi-Agent B2B Sales Automation Platform *client work @ Maxflow AI* *2026*

- Master routing agent delegating to specialist agents for partner research, a **100-point lead-scoring engine** (pure Python), email drafting, controlled batch delivery, and campaign analytics; CRM integration through a custom MCP bridge and a Cloud Run webhook that converts CRM events into agent runs. **Reduced a 2–3 hour research-and-quote cycle to about three minutes.**

Inventory Intelligence Decision Engine *client work @ Maxflow AI* *2026*

- Python-first engine computing days-of-inventory, demand trends, and warehouse-transfer recommendations with **auditable reason codes**. The LLM never calculates — it writes the executive summary from computed JSON. A hallucination-proof architecture for decisions that move real stock.

Cuffless Blood Pressure Monitoring — Sensor to App *github.com/BuccKyy* *2024 – 2025*

- Full hardware-to-model stack: **STM32 + MAX30102** firmware, Bluetooth streaming, CNN-LSTM and Transformer training on MIMIC-III, TensorFlow Lite on-device inference. Selected for the **Biomedical Engineering Innovation Competition 2025**; **90% user satisfaction** across 20+ test participants as Team Lead
- Companion **Android app** (Kotlin, Camera2 API): two-stage TF Lite network with 85 MB models, 1024-point sliding-window buffering with signal normalisation, background-thread memory management, and Google Drive cloud sync via OAuth

LLM Video Translation Pipeline & **RAG Claim-Checker** *github.com/BuccKyy* *2025*

- Generative-AI pipeline (Whisper ASR → LLM translation → TTS → FFmpeg remux) with latency-aware, independently testable stages, Dockerised and hardened against API rate limits; plus a deterministic **BM25 claim-checker** that grounds every generated claim in source evidence as a re-runnable regression gate for RAG features

EDUCATION

University of Science, VNU-HCM Ho Chi Minh City, Vietnam *2021 – 2025*
Bachelor of Engineering, Electronics and Telecommunication Engineering GPA: 3.4 / 4.0

Relevant coursework: Machine Learning, Deep Learning, Digital Signal Processing, Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Embedded Systems, Software Testing & Verification

TECHNICAL SKILLS

Agentic AI & LLMs	Multi-agent orchestration (LangGraph, n8n), **MCP**, tool-calling & function schemas, memory & context handling, guardrails & fallback logic, human-in-the-loop review, prompt engineering, **Claude Code**; Anthropic Claude, OpenAI GPT, LLaMA, Qwen (Ollama), Whisper, Hugging Face
Retrieval & RAG	RAG pipelines, embeddings, semantic search, **hybrid BM25 + dense retrieval**, vector databases (FAISS, Chroma, PGVector), grounding & claim verification
Machine Learning	**PyTorch**, TensorFlow/Keras, scikit-learn; CNN, LSTM, Transformer, attention mechanisms; training, fine-tuning, custom loss functions, evaluation protocols (MAE/RMSE), quantised on-device inference (TF Lite), digital signal processing
Backend & Data	**Python** (advanced), FastAPI, REST API design, event-driven triggers, webhooks, jobs & retries; SQL / relational schema design (PostgreSQL, MySQL, SQLite), ETL & data pipelines, Pandas, NumPy
Inference & Cost	Model routing for cost/latency, context & prompt optimisation, caching, response streaming, timeout / retry / rate-limit handling, production monitoring
QA, DevOps & Tools	**Playwright**, Selenium, deterministic LLM output validation, regression & edge-case testing, transcript-based debugging; **Docker**, Cloud Run / GCP, CI/CD, Git, Linux, structured logging & observability, FFmpeg, OpenCV
Embedded & Mobile	C/C++, STM32, MAX30102, GPIO / I2C / UART, Bluetooth; Kotlin, Android (Camera2 API); also Java, MATLAB, Verilog, JavaScript / HTML

AWARDS & CERTIFICATIONS

- **VPBank Technology Hackathon 2025** — Senior Track participant | **Biomedical Engineering Innovation Competition 2025** — Certificate; Team Lead, IoT Blood Pressure Monitoring System
- **Machine Learning with Python** — Coursera | **Deep Learning & Neural Networks with Keras** — Coursera | **Building Deep Learning Models with Keras and TensorFlow** — Coursera
- **Languages:** Vietnamese (native) | English (professional working proficiency — technical documentation, async remote collaboration)